Project Proposal

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Agentic AI

Agent: Meal Planner AI Agent

1. Problem statement & motivation
    - What real-world business problem are you addressing?
        i. Many people struggle to find healthy, nutrient dense, and affordable meal ideas that accurately align with their goals and budget. Usually, there is an overwhelming mental load that is associated with healthy eating and meal planning, and this agent is designed to solve that. Current apps often get too complicated, straying from simple methods and instead requiring users to input every caloric intake, push one goal over another, and often fail to take into consideration what is in your fridge or pantry. This quickly leads into a snowball effect of making unhealthy eating habits, wasting food, and overspending. This agent can assist to get rid of the anxiousness that clouds healthy eating and meal planning and can turn something that is usually extensive and unfulfilling to be enjoyable.
    - Who are the target users?
        i. College students who have limited schedules and resources
        ii. Young professionals with newfound independence and a lack of time
        iii. Those with specific dietary restrictions, intolerances, or preferences
        iv. Budget conscious individuals looking to make the most with what they have
    - Why is an *agentic* AI solution appropriate here (vs. a human or an external party defines the workflow)?

An agentic AI solution is appropriate here because it is a more efficient way to accomplish the tasks listed previously. While hiring a private chef or dietitian can provide similar care and human interaction, it can easily get very expensive, ineffective, and cannot easily adjust to unpredictable and ever changing schedules that young adults and professionals often have. This agent can fill that gap by providing an extensive set of information that the user requires, and can be easily adjusted based on the user's needs at that time. Instead of giving you a list and instructions to follow, this agent can work with you whenever you need and can cross reference with whatever data provided. Doing this can allow you to create a simpler routine for yourself while still being aligned with your goals.

2. Use cases & scenarios
    - 2–3 concrete user scenarios (e.g., "An user wants to…").

        User Scenario #1: Weekly Meal Planning

        An user wants to plan their meals for the week based on their work schedule.

        Agent tasks include:

        -Collect user's dietary preferences/restrictions

        -Ask for user's week schedule

        -Analyze user's goals

        -Generate a meal plan

        -Generate grocery list for meal plan

        User Scenario #2: Pantry Meals

        An user wants to cook a meal with only the items they currently have.

        Agent Tasks:

-Collect user’s pantry items

-Ask user’s meal preferences/counts

-Give user recipe options

-Offer substitutes and changes

3. Initial system design
    - Preliminary list of agents and their roles (e.g., Visualization agent, Planner, Data Fetcher, Summarizer, Critic..).
        i. Inventory Tracking agent: Can manage a real time database of everything in the user’s pantry/fridge based on their inputs. The data will also include specific quantities so the user will be aware when meal planning.
        ii. Dietician agent: Acts as an information database for nutrition, can evaluate what the user needs depending on their specific restrictions and preferences. This will also align with any specific health goals and provide recommendations.
        iii. Recipe fetcher agent: Goes through different online databases, including but not limited to blogs, recipe books, and other websites to find a variety of options preferable to the user.
        iv. Resource optimizer agent: Integrates the inventory tracking agent along with the recipe fetcher agent to ensure that all of the user’s resources(pantry/fridge items) are being used properly and in a timely manner to minimize waste.
        v. Weekly plan Summarizer agent: Compiles all of the information input by the user to create an easily readable summary of the meal plan, specifying

daily meals, quantities, resources used and left, and specific nutrition information if requested, as well as a shopping list.

  vi. Grocery store connector agent: Optional link to grocery stores to see what products are available where and eventually will be able to facilitate online ordering.

- Expected tools/APIs or data sources (e.g., web search, Google Docs, Notion, Figma, internal API, local files).
  i. Web search
  ii. Google docs
  iii. Grocery store data
  iv. Nutrition data
  v. Recipe data

4. Feasibility & risks
- Potential challenges (e.g., hallucination, long context, tool failures).
  i. Extensive and overwhelming amount of data
  ii. Lack of sufficient API's
  iii. Repetitive meal structures for more specific dietary needed
- Any constraints (e.g., rate limits, privacy constraints, computation).
  i. Insufficient user input data
  ii. Dietary information data gaps
  iii. Non-medical expertise
- Time line for project execution (cannot extend beyond the class duration).
  i. Project milestone 1 due 03/05

1. Start working on front end by 02/15
2. Finish midway by 02/20
3. Get it done 02/26,
4. ask questions/feedback 02/27

ii. Project milestone 2 due 03/12

1. Start building multi agent workflow by 03/6
2. Finish by 03/10
3. Ask any last minute questions and submit by 03/12

iii. Project milestone 3: due 03/26

1. Start wrapping it up by 03/20
2. Ask last minute questions by 03/23
3. Submit by 03/26